PRASUN JHA

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Third-year Mathematics & Computing undergraduate focused on robotics, reinforcement learning, autonomous navigation and control. Builds end-to-end robotic systems spanning perception, decision-making, simulation and embedded hardware.

Education

Birla Institute of Technology, Mesra Aug 2024 – Jul 2029
Integrated BS-MS in Mathematics and Computing **CGPA: 8.1/10**
Coursework: Linear Algebra, Probability, Differential Equations, Optimization, Data Structures & Algorithms, OOP

Research & Technical Projects

Vision-Guided Socially Aware Navigation — Ongoing Research [GitHub] 2026 – Present
- Built a **camera-only** assistive navigation stack in **ROS 2 Humble** and Gazebo Fortress: an Ackermann-drive robot in a simulated hospital corridor with hazard detection and teleoperation, using **no LiDAR or depth sensing**.
- Simulated crowds with **ORCA (RVO2)** pedestrian nodes and added an Open-RMF Airport Terminal crowd stress-test.
- Developing a **PPO** baseline (Stable-Baselines3) on camera/OpenCV features and testing **VLM-derived scene semantics** as observations or reward shaping, evaluated on proxemics violations, collisions and time-to-goal.

Modular Nav2 Navigation Stack — ERIC Robotics Technical Assignment [GitHub] Oct 2026
- Built Nav2 from individual plugins (map_server, **AMCL**, NavFn planner, **DWB** controller, BT navigator) with separate lifecycle managers for a differential-drive robot in ROS 2 Humble and Gazebo.
- Found and fixed **10 bugs** in the starter code, including a 1.5 m LiDAR range that blocked localization and a wheel-separation error recalibrated from a measured turn; AMCL error **0.02–0.26 m** vs. ground truth.
- Traced missed goals to **/tf flooding** starving the controller (∼330 → 116 msg/s); reached **7/8 goals** within 0.21 m.

Fuzzy Robot Coordinator — Multi-Robot Task Allocation [GitHub] 2026
- Built an **explainable** allocator that scores robot–task suitability from battery, distance, workload, payload ratio and urgency through **eight Sugeno-style fuzzy rules**, compared against a weighted mathematical baseline.
- Solved **exact one-to-one assignment** maximizing urgency-weighted suitability under payload limits.
- Integrated the planner with a **MuJoCo** warehouse of **three differential-drive robots** with autonomous job execution and **battery-aware charging** (suspend at 25%, recharge to 85%); plus a web dashboard and **17 tests**.

RL Self-Balancing Motorcycle — ROS & Gazebo [GitHub] Jun 2025 – Aug 2025
- Trained a **Soft Actor-Critic (SAC)** agent to self-stabilize a dynamically unstable two-wheeled motorcycle and navigate it to target waypoints in Gazebo 11 with ROS Noetic.
- Built a **custom Gym environment** bridging Stable-Baselines3 and Gazebo via ROS topics/services, run in Docker.

TortoiseBot Navigation Stack — ROS 2 [GitHub] 2026 – Present
- Architecting a **distributed ROS 2 stack** for the RigBetel Tortoise robot across a Raspberry Pi 4 edge node (micro-ROS, LiDAR, camera) and a Pi 5 Docker compute node, with Gazebo simulation and **SLAM Toolbox**/Nav2.

Flight Simulation Control Systems — JSBSim [GitHub] Apr 2026 – Present
- Built a **real-time flight-dynamics pipeline** in JSBSim (Cessna 172P, custom RC plane) with FlightGear over UDP, and tuned **PID controllers** for stable flight under disturbances.

Inverted Pendulum on a 4-Wheel Cart — Embedded Control [GitHub] 2024
- Implemented real-time balancing with an ESP8266, MPU6050 IMU, **Kalman-filter** angle fusion and **PID control** driving four geared DC motors through an L298N driver.
- Built a Wi-Fi web interface for **live PID tuning** without reflashing firmware.

Technical Skills

Robotics	ROS 2, Gazebo, MuJoCo, Nav2, AMCL, SLAM Toolbox, TF2, micro-ROS, CoppeliaSim, ORCA
ML / RL / CV	PyTorch, Stable-Baselines3, PPO, SAC, Q-learning, Gymnasium, OpenCV, YOLOv8, VLMs
Control	PID, Kalman Filtering, Fuzzy Logic (Sugeno), Task Allocation, Optimization
Embedded	ESP32, ESP8266, Arduino, Raspberry Pi, Jetson Orin Nano
Languages & Tools	Python, C++, C, Linux, Docker, Git

Achievements

- **Zonal Runner-up**, Ramakrishna Mission Innovation Challenge — smart assistive eyeglasses built on Raspberry Pi.
- **Zonal Runner-up**, Dassault Systèmes Innovation Program — foldable 3D-printed helmet design.
- **High Commendation**, Sokovia Accords MUN; **2nd Place**, Pantheon Quiz; **Special Mention**, Online MUN.

Leadership & Open Source

- **General Secretary, Robolution** (Robotics Club, BIT Mesra; May 2026 – Present) — leading project architecture, technical coordination, sponsorship and junior training; earlier Intern and Embedded Member (2025–26).
- **GSoC 2026 Proposals, JdeRobot** — C++ exercise extension; deep-learning power-tower inspection tool.
- **Events Coordinator, UNESQUO**; **Treasurer, SPIC MACAY**; BITMUN 26 design lead; as school council member, organized city fest INFINITY (**1,500+ participants**).